

Integration of subcritical water pretreatment and anaerobic digestion technologies for valorization of açai processing industries residues

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ABSTRACT

The objective of this work was to study the valorization of residual biomass and wastewater from açai fruit processing to obtain energy products and higher added value from such residues, through the integration between subcritical waterpretreatment and anaerobic digestion technologies. Subcritical watertechnology was performed to pretreat the residual biomass. For this process, the reactorwas feed with 5 g of açai residue and a solvent to biomass mass ratio of 80 g of water per gram of açai residue was used. The process was carried out at 170 and 200 °C, during 20 min, using water flow of 20 mL min⁻¹ (15 Mpa system). Then, the residual biomass was submitted to the anaerobic digestion process using a reactor with pretreatment (integration) and a control reactor, both in semi-continuous regime and under mesophilic conditions (35 °C). The açai stone had a high carbohydrate concentration including 43.81% cellulose and 25.89% hemicellulose. For the evaluated conditions, subcritical water pretreatment at 200 °C gave the bestresultsforyield of reducing sugars, total reducing sugars and monosaccharide concentration. Anaerobic digestion experiments resulted in analkalinity between 150 and 1537 mgCaCO₃.L⁻¹ for the control reactor and between 249 and 491 mgCaCO₃.L⁻¹ for the reactor integration.Acetic acid was the main volatile fatty acid produced in both reactors, but butyric acid also had a very high concentration in the control reactor. The methane composition in the biogas for the integration reactor in the stabilization phase was between 60 and 80%, whilefor the control reactor it was in the range between 40 and 60%.The accumulated biogas yield was 7.79 and 791.81 L.KgTVS⁻¹ for the control and integration reactor, respectively. Anaerobic digestion using the açai hydrolysate obtained by subcritical water pretreatment showed a significant advantage over the control digestion, resulting in30% more methane in the biogas and 100 times higheryield.

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1. Introduction

Fossil fuels are the primary source of energy today. However, its burning generates high levels of atmospheric pollution, emission of toxic substances and greenhouse gases, the principal responsible for global warming and current climate changes (Naran et al., 2016). Besides, fossil fuels are poorly distributed throughout the world and may be deplete soon (Deepanraj et al., 2017). This increases the interest in renewable fuels that can replace conventional energy sources, such as biofuels derived from biomass (Grosser and Neczaj, 2016; Kanchanasuta and Pisutpaisal, 2016).

Biomass is all solid carbon matter produced by living beings

(animals, plants, fungi, and protists) that can be used as fuel and/or energy. The biomass energy comes from the photosynthesis process and is stored in the form of carbohydrates (mainly as cellulose, hemicellulose) (Karayildirim et al., 2008), the chemical oxidation is also another source of energy for biomass like for nitrifies (Sepehri and Sarrafzadeh, 2018). In the food industry, specifically in the processing of açai fruit (*Euterpe oleracea*Martius) whose fruit has a low yield of pulp (part used in processing), on average only 10% in mass, the remainder is residual biomass (seeds and fibers). Currently, In Brazil, there is no large-scale recovery of this solid waste with high-energy potential (Matos et al., 2011; Virmond et al., 2012; Yuyama et al., 2011).

The energy exploitation of lignocellulosic biomass or solid waste from the food industry can be achieved through anaerobic digestion technology with the production of biogas. The process consists of the conversion of organic matter into biogas through the action of

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different groups of microorganisms through the steps of hydrolysis, acidogenesis, acetogenesis, and methanogenesis. The biogas used in the elaboration of fuel for vehicles, electric and thermal energy (Deepanraj et al., 2014; Risberg et al., 2013; Zhang et al., 2014). However, in the anaerobic digestion of lignocellulosic residues, hydrolysis can be a limiting stage given that the polymeric structure of the material makes it difficult to convert to more biodegradable molecules (Hendriks and Zeeman, 2009; Martinez-Jimenez et al., 2017; Pinto et al., 2018). The use of pre-treatment techniques, such as mechanical, thermal, chemical or biological treatments, can then be considered (Kim, 2013). Conventional methods of acid hydrolysis, vapor explosions, and enzymatic hydrolysis have been extensively studied in recent years, but they all have several disadvantages to the environment. Currently, hydrothermal technology, especially the sub/supercritical water hydrolysis processes have shown excellent results of energy recovery and/or biofuel production from lignocellulosic residues, presenting a very high yield compared to conventional technologies, in addition to and to be considered sustainable and environmentally friendly (Escamilla-Alvarado et al., 2014; Matsakas et al., 2015; Prado et al., 2016, 2014).

The water in the sub/supercritical state becomes an alternative solvent for the dissolution of the biomass due to its low viscosity and high diffusivity that facilitates its penetration into the complex structure of the lignocellulosic matrix, while its low dielectric constant, similar to organic solvents (non-polar), increases the solubility of organic compounds. The physical properties of the water as density, ionic product, and dielectric constant can be refined by different temperature and pressure conditions, making the hydrolysis of the biomass fractions a fast and selective process without causing environmental or operational problems (Cocero et al., 2018). Several studies have been carried out with hydrothermal technology using lignocellulosic residues such as corn stalks (Zhao et al., 2012), rice bran (Sunphorka et al., 2012), sugarcane bagasse (Lachos-Perez et al., 2016; Zhu et al., 2013), and coffee powder (Mayanga-Torres et al., 2017).

Further, studies are needed to optimize process parameters or to improve equipment design, as hydrolysis rates and yields depend on the intrinsic characteristics of the residue (composition, cell wall structure, monosaccharides, type of bonds between them, lignin). Currently, in the literature, there are few studies of pretreatment of lignocellulosic residues for the application of their hydrolysates in other processes such as in anaerobic digesters. To evaluate the integration of hydrothermal and anaerobic digestion technologies a system is proposed with the use of the hydrolysis process with subcritical water as a pretreatment to obtain a final product (fermentable sugars) that will be used in anaerobic digestion process for the production of a methane rich biogas.

2. Materials and methods

2.1. Raw material and its preparation

Solid residues (açai seeds) and wastewater from açai fruit processing, was provided by the Villa Roxa 'Açai e alimentação saudável'. The mesophilic inoculum was supplied by São Martinho S.A, a group from the sugarcane area.

The açai seeds were oven dried (Fenem, model 315 SE) at 105 °C for 8 h, then ground in a mill (Marconi, model MA 340) equipped with a 1 mm sieve, finally packed in a bag plastic and stored in a freezer at −18 °C for later use in the experiments of subcritical water pretreatment and anaerobic digestion (Fig. 1).

2.2. Experimental setup for subcritical water pretreatment

The experiments of subcritical water pretreatment of açai

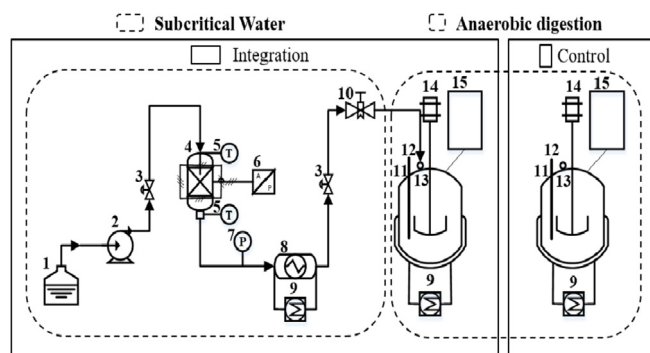


Fig. 1. Schematic diagram for the combined process of subcritical hydrolysis with Anaerobic Digestion. (1) Water tank, (2) HPLC pump, (3) Block valves, (4) Subcritical reactor (5) Thermocouples, (6) temperature controller, (7) Manometer, (8) Heat exchanger, (9) Refrigeration bath (10) Micrometer valve, (11) Bioreactors, (12) Tubes for sample collection, (13) Holes for Alimentation and pH control, (14) Stirrers, and (15) Bags for Biogas collect.

residue were performed in a semi-continuous flow unit (i.e. batch to solid and continuous to liquid). The equipment contains a stainless steel hydrolysis reactor (with an internal volume of 110 mL), supporting the maximum working conditions of 400 °C and 42 MPa. The reactor is heated by an electric system (1500 W) jacket type insulated by ceramic fiber. The water used in the hydrolysis process was pumped by a high-pressure pump (double piston pump, model Prep 36 Pump). The pressure in the system was controlled by a micrometer valve (Autoclave Engineers, model 10VRMM2812) and measured by a pressure gauge. The hydrolysis temperature was monitored by thermocouples (type K) inside and at the exit of the reactor. The hot hydrolysate exiting the reactor was cooled to below 30 °C in a shell-and-tube stainless steel heat exchanger coupled to a thermostatic bath (Marconi, model MA-184) operating at 10 °C using water as a coolant.

Previous work carried out in the BIOTAR laboratory indicates the best operational conditions for the hydrolysis process of lignocellulosic biomasses using subcritical water. Lachos-Perez et al. (2016) worked with sugarcane bagasse, Lachos-Perez et al. (2017) with sugarcane straw, Lachos-Perez et al. (2018) with orange peel, Mayanga-Torres et al. (2017) with coffee residues, powder and defatted cake, and Torres-mayanga et al. (2019) with malt bagasse. Based on the previous results, in this present work, we chose to use 5.0 g of açai residue in the hydrolysis reactor and maintain the water flow and pressure parameters constant at 20 mL min^{−1} and 15 MPa respectively. The mass ratio of solvent to biomass (S.B^{−1}) was 80 g of water per g of açai seed. The temperature parameter was evaluated in two conditions, 170 and 200 °C, in duplicate; since this is the most crucial operational parameter and that it varies according to the type of biomass studied. The hydrolysis time for both conditions evaluated was 20 min, and after this period, the hydrolysates obtained were analyzed according to pH values, reducing sugars, total reducing sugars, the composition of sugars, organic acids and inhibitors.

Allied to temperature the residence time (t_R) of the solvent with the biomass and severity factor (R_0) has the significant influence on the hydrolysis yield of lignocellulosic biomasses (Getachew and Chun, 2017). Thus, they were determined using equations (1) and (2).

$$t_R = \frac{V_R \cdot \rho_R(T, P)}{Q \cdot \rho_0} \quad (1)$$

Where " V_R " is the volume of the reactor (m³); " Q " is the volumetric flow rate (m³.min^{−1}); " ρ_0 " is the density of the feed water (Kg.m^{−3})

and " ρ_R " is the density of water under temperature and pressure process (Kg.m^{-3}), which was estimated using the Book Chemical, Biochemical, and Engineering Thermodynamics (Sandler, 2017).

$$R_0 = t_R \cdot e^{\left(\frac{T-100}{14.75}\right)} \quad (2)$$

Where " t_R " is the residence time (min), " T " is temperature ($^{\circ}\text{C}$), "100" is the reference temperature and "14.74" is an empirical parameter related to the activation energy, assuming a kinetic of the first order. The values were expressed in logarithmic scale.

2.3. Experimental setup for anaerobic digestion

Two anaerobic digestion reactors were performed. The first was called of control reactor (açai seed and wastewater of açai processing), without pretreatment of açai seed. The configuration was high solids content (25% of total solids and 75% of liquid, where 30% of the liquid was mesophilic sludge, and the remaining was wastewater of açai processing). The second reactor was called integration reactor (with pretreatment) because the hydrolyzed of the açaiseed obtained from the subcritical system was used to anaerobic digestion reactor with low solids configuration, having 70% hydrolyzed and 30% mesophilic sludge. The flow rate was 300 mL per week in 2.15 L of waste inside the reactor then TDH value was 35 days.

The experiments were carried out in a semi-continuous regime, with sample retention and substrate feeding three times a week, thus maintaining the constant volume. The digestion was carried out in standard stainless steel reactors, of the stirred tank type, with a useful volume of 4.3 L. Maintained under mesophilic conditions of temperature (35°C) with the aid of a thermostatic bath (Tecnal, model TE-2005). The working volume of each reactor was 50% of the total volume, and the other 50% was destined for foam and biogas produced. The experiments had a hydraulic retention time (HRT) of 35 days since 100 mL of the reaction mixture contained in the reactors was withdrawn and 100 mL of substrate (25% solids and 75% wastewater of açai processing) was added to the control reactor, and 100% of hydrolyzed to the reactor integration).

The homogeneity of the contents inside the reactors was maintained through a system of mechanical stirrers (Fisatom®, model 715), used for 10 min prior to sample collection and 10 min after addition of substrate or hydroxide for pH control.

Anaerobic digestion had a duration of 72 days. During the first 10 days of the experiment, no pH control was done to favor the initial stages (hydrolysis and acidogenesis) of the digestion process. After this time the pH of the reactors were maintained between 7 and 8 with the addition of 6N sodium hydroxide (NaOH). Sample collection, substrate feed, base addition for pH control and biogas outlet to the gas collection bag (Supelco Analytical) occurred through the holes in the reactor lids.

2.4. Analytical methods

The analytical methods used in this work were: Density was determined by the ratio of mass to volume similarly using a graduated cylinder to the liquid and solid raw materials (César et al., 2004). pH (4500-H⁺ B), Total solids and moisture (2540B), Total fixed and volatile solids (2540E), Alkalinity (2320 B), Ammonium nitrogen (45000-NH₃ C), Chemical oxygen demand (4520 D) Total nitrogen (4500-N_{org}B), all according to the Standard Methods for the Examination of Water and Wastewater (APHA, 2017). Proteins (960.52) determined by multiplying the total nitrogen content by the factor 6.25, Lipids (920.39) both according to the Official

Methods of Analysis of the Association Analytical Chemists (AOAC, 2016). Total Extractives, Extractives in water, and Extractives in ethanol all according to the Technical Report NREL/TP-510-42619 (Sluiter et al., 2008), Total Lignin, Lignin insoluble, and Lignin soluble all according to the Technical Report NREL/TP-510-42618 (Sluiter et al., 2012). Holocellulose, Cellulose, and Hemicellulose (Wise et al., 1946). Total reducing sugar and Reducing sugar content were determined using the colorimetric method proposed by Somogyi and modified by Nelson (1944).

Determination of the concentration of sugars and inhibitors was done by High Performance Liquid Chromatography (HPLC) using a Dionex Ultimate 3000 system (Dionex Softron GmbH, Germany) equipped with a Bio-Rad Aminex HPX-87H column (300 mm $\times$ 7.8 mm) at 60°C and 5.0 mM sulfuric acid as mobile phase at a flow rate of 0.6 mL min^{-1} . Sugars were detected using an IR detector (Shodex RI-101), whereas for the detection of furfural and 5-HMF a UV detector at 254 nm was used.

To determination of the volatile fatty acids, the samples were diluted with distilled water according to the concentration of each sample and acidified with 3M phosphoric acid in the proportion of 1:20. The determination was performed using a gas chromatograph (GC) (Shimadzu®, model GC 2014) and capillary chromatographic column (Agilent Technologies®, model DB-WAX) with 30 m in length, 0.25 mm internal diameter and 0.25 μm film thickness which allows separation of the components present in the sample according to their interaction with the stationary phase. The operational parameters were follows: Temperatures of the injection port and detector were both set to 250°C ; initial temperature of GC column was 100°C (held for 1 min), and then increased by $8^{\circ}\text{C.min}^{-1}$ to 200°C (held for 2 min); Injected sample volume was 1 μL ; No Split; Nitrogen (N₂) was used as a carrier gas with a rate of 1 mL min^{-1} and pressure of 92.4 KPa. The quantification of acetic, propionic, isobutyric, butyric, isovaleric, valeric and hexanoic acids was performed using standard calibration curves (Sigma-Aldrich).

The biogas was measured through your collection in gas sampling bags (Supelco Analytical) coupled with bioreactors. The cumulative volume and cumulative yield of biogas were calculated according to Equations (1) and (2), respectively. Where, "Y" is yield, "V" is volume, "gTVS" is the amount of Total Volatile Solids added in the reactors and "n" is the number of days.

$$V_{\text{accumulated}}(L) = \sum_{n=1}^n V_n \quad (3)$$

$$Y_{\text{accumulated}}\left(\frac{L}{\text{gTVS}}\right) = \sum_{n=1}^n \frac{V_n}{\text{gTVS}} \quad (4)$$

The composition of the biogas produced by the integration and control reactors was determined by direct biogas collation in the reactors and injection manually into a gas chromatograph (Shimadzu®, model GC 2014) equipped with a thermal conductivity detector (TCD) and a micro-packed column for compound separation (ShinCarbon, ST 50/80 mesh) of 6m in length, 3 mm of internal diameter. To determine biogas composition (H₂, CH₄, and CO₂), the following chromatographic conditions were used: temperatures of the injection port and detector were both set to 200°C ; initial temperature of GC column was 50°C (held for 3 min), and then increased by $5^{\circ}\text{C.min}^{-1}$ to 180°C (held for 5 min); the sample volume injected was 0.5 mL; and N₂ was used as a carrier gas (35 mL min^{-1} , 5 bar).

3. Results and discussion

3.1. Physical-chemical characterization of raw materials

Table 1 shows the results of the physicochemical characterization of the residues used in this work. For the determination of the centesimal composition of the biomass of the açai stone, the moisture content and the total fixed solids of the material were determined previously, which were 11.49 and 1.18%, respectively. These results show values higher than those of 8.1% of humidity and lower than the total fixed solids of 3.5% studied by Oliveira (2014). However, some studies in the literature show that these types of materials often present a high silica content (Seye et al., 2000), so perhaps the difference in solids values fixed. Concerning the moisture content, it can vary greatly, mainly due to storage conditions.

The component of the biomass of the açai berry found in the highest quantity was cellulose with a value of 43.81%, followed by hemicellulose with 25.89% and lignin with 24.56%. Being 22.99% insoluble lignin and only 1.57% soluble lignin, that is, insoluble lignin represents 93.6% of total lignin. Concerning the percentage of total carbohydrates (cellulose and hemicellulose), the content of 69.70% was observed, relatively close to 63.45 and 65.46% reported by Oliveira (2014) and Rodríguez-Zúñiga et al. (2008). From this high carbohydrate content obtained in the characterization of the açai residue is justified its use for biogas production. However, the total nitrogen value, which is an essential nutrient for the microorganisms of the anaerobic digestion, was relatively low (less than 1%). "The açai seeds presented high concentration of carbohydrates (carbon source) and low nitrogen, resulting in a high C/N ratio (>35). The C/N parameter indicates the ratio between Carbon (C) and Nitrogen (N) in biomass composition, and the optimal C/N ratio for anaerobic digestion was between 15 and 45. Higher values (>45) decrease the rate of reaction, while smaller amounts (<15) cause inhibition by excess of nitrogen. The carbon is used as a substrate by microorganisms and nitrogen as nutrient source. Teixeira et al. (2005) found in their study a ratio of 48, moderately above the maximum of 45 recommended by Wagner et al. (2013).

Another relevant point for application of açai residues in the anaerobic digestion process is the concentration of lignin, which can strongly affect the biodegradability of the material, making it

difficult to convert the lignocellulosic biomass to biogas, besides that, is responsible for the generation of toxic phenolic compounds and inhibitors (Hagos et al., 2017). Although the lignin content of 24.56% of the açai stone is considered reasonably high, it is in the range of the main lignocellulosic residues produced in the country.

The density of the raw materials were 1,151, 1,023 and 996 kg m⁻³ for the açai residue, mesophilic sludge and wastewater, respectively. The last value is very close to the 997 kg m⁻³ which are the value for pure water at 25 °C (Sandler, 2017), due to the low total solids concentration, only 1.21%, which was constituted mainly by total volatile solids with 1.14%. In addition, the mesophilic sludge presented a higher density than water, possibly because it had 10.44% of total solids, considered high for a sludge, since the work of Liu et al. (2018), Náthia-Neves et al. (2018), and Pinto et al. (2018) with this type of mud found 4.70, 3.14, and 2.47%, respectively. The ash content of this study was also high, with 2.80%, representing approximately 27% of the total solids content.

Another component that the biomass presented in high concentration was the total nitrogen, with 8.39% in dry base, something extremely positive, for the digestion of the residue of açai that contains low concentrations of this chemical element. It was not possible to determine the protein content for the sludge, since its total nitrogen value is the result of the decomposition of the previously treated residues, that is, it does not have a protein structure organized and quantifiable by means of a conversion factor.

3.2. Subcritical water pretreatment

3.2.1. Reducing sugars and total reducing sugars in the hydrolysate

Fig. 2 shows the release of sugars from the residual biomass of açai according to the time to each evaluated temperature (170 and 200 °C). Graphs (A) and (C) are expressed as gram equivalents of glucose per liter of hydrolysate, useful to know the sugar concentration that will be added in the anaerobic digestion reactor. Graphs (B) and (D), expressed as gram equivalents of glucose per 100 g of dry açai residue are important in visualizing the yield of sugars released concerning the amount of residual biomass.

Graphs (A) and (C) show similar behavior, i.e., an increase in sugars in the first 4 min and then a decrease in the concentration of hydrolyzed sugars. In this way, it observes a higher level of sugars in the second point of kinetics, at 4 min, then the concentration of

Table 1
Physical-chemical composition of raw materials.

Parameters	Açai residue	Inoculum	Wastewater
Density (kg/m ³)	1151.06 ± 46.81	1023.16 ± 1.60	996.23 ± 9.81
pH	6.11 ± 0.03	8.42 ± 0.05	7.00 ± 0.02
Moisture (%)	11.49 ± 0.10	89.56 ± 0.22	98.79 ± 0.08
Total solid (%)	88.51 ± 0.10	10.44 ± 0.22	1.21 ± 0.08
Total fixed solid (%)	1.18 ± 0.04	2.80 ± 0.07	0.07 ± 0.01
Total volatile solid (%)	87.33 ± 0.13	7.64 ± 0.29	1.14 ± 0.08
Alkalinity (mgCaCO ₃ .L ⁻¹)	30.37 ± 2.77	914.34 ± 14.65	20.78 ± 2.77
Ammonium nitrogen (mgNH ₃ .L ⁻¹)	54.44 ± 6.74	2289.33 ± 58.70	24.67 ± 3.29
Chemical oxygen demand (gCOD.L ⁻¹)	72.57 ± 1.49	59.36 ± 1.77	13.37 ± 0.25
Dry basis			
Total nitrogen (%)	0.84 ± 0.02	8.39 ± 0.28	—
Proteins (%)	5.27 ± 0.14	—	—
Lipids (%)	3.33 ± 0.31	—	—
Total Extractives (%)	7.71 ± 0.06	—	—
Extractives in water (%)	5.32 ± 0.20	—	—
Extractives in ethanol (%)	2.39 ± 0.12	—	—
Hollocelulose (%)	69.70 ± 0.90	—	—
Cellulose (%)	43.81 ± 3.39	—	—
Hemicellulose (%)	25.89 ± 2.49	—	—
Total Lignin (%)	24.56 ± 0.58	—	—
Lignin insoluble (%)	22.99 ± 0.47	—	—
Lignin soluble (%)	1.57 ± 0.12	—	—

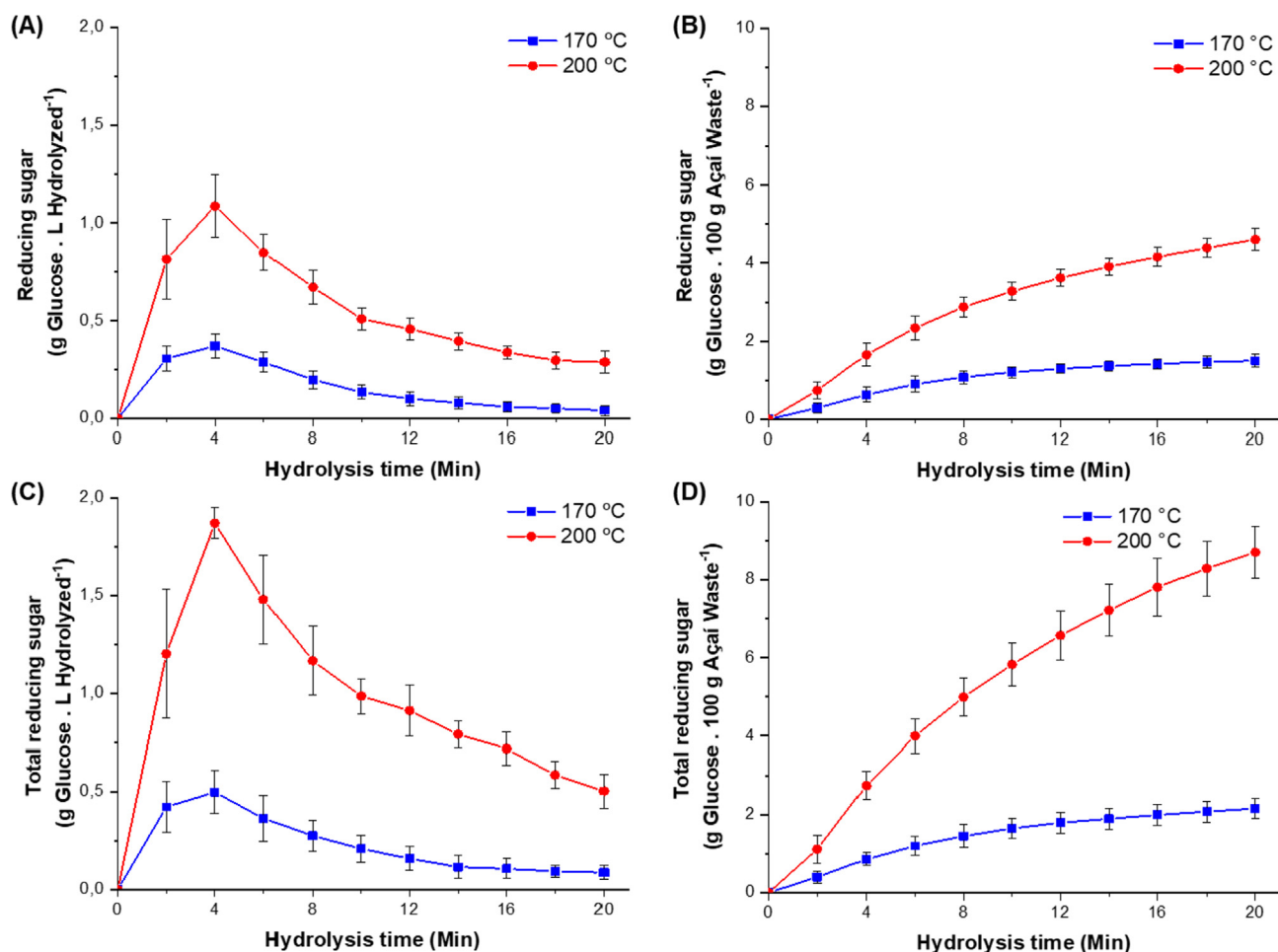


Fig. 2. Yield of reducing sugars and total reducing sugars at temperatures of 170 °C and 200 °C. (A) Reducing sugars content in hydrolyzates (B) Reducing sugar content due to açai residue, (C) Total reducing sugars content in hydrolyzates, and (D) Total reducing sugar content due to açai residue.

sugars decreases in both temperatures analyzed. For the reducing sugars, the maximum levels were 0.37, and 1.09 gL⁻¹ and the averages were 0.16 and 0.57 gL⁻¹ for the temperatures of 170 and 200 °C, respectively, these average concentrations of the hydrolysates had an increase of 43.7 and 78.9% in total reducing sugars, resulting in values of 0.23 and 1.02 gL⁻¹. As for the yields shown in graphs (B) and (D), the increase concerning the total reducing and reducing sugars were 44.0 and 88.7% for the temperatures of 170 and 200 °C respectively, 1.05 and 4.61 to 2.16 and 8.70 g.100 g⁻¹. This yield of total reducing sugars at 200 °C was higher than that reported by Mayanga-Torres et al. (2017) whose work used coffee residue.

These data generally reveal an increase in sugar yield as the hydrolysis temperature is increased, as expected (Lü and Saka, 2010; Toor et al., 2011). These results corroborate with literature reports that the reaction temperature is the most important parameter of the hydrolysis process in subcritical water (Cardenas-toro et al., 2014; Ciftci and Saldaña, 2015). The increase in the sugar yield obtained with the increase in temperature can be mainly attributed to the thermal effects on the kinetics of hydrolysis (Abdelmoez et al., 2014), and secondarily to the increase of the autoionization constant of the water, the constant ionization of H⁺ and OH⁻ ions concentrations in the reaction mixture, thus facilitating hydrolysis catalysed by acids formed from cellulose and hemicellulose (Zhu et al., 2013).

3.2.2. Chemical composition of the hydrolysates

Table 2 shows the results of the chemical composition of the hydrolysates produced by subcritical water pretreatment, as well as their pH values and amount of residual solid remaining in the reactor. As can be seen, the temperature increase had a positive effect on the release of sugars and in the production of ethanol and organic acids. A possible explanation for this is the influence of the pH of the reaction medium, more specifically on the concentration of acetic acid, because as pointed out by (Costa et al., 2014), the acetic acid released by the cleavage of the acetyl groups can act as a catalyst for hydrolysis reaction of the hemicellulose, contributing to the increase of the sugar release.

Given these results, it can be inferred that the increase in temperature contributed to the hydrolysis of hemicellulose, leading to an increase in the release of C5 (xylose) and C6 (mannose and galactose) sugars, because hemicellulose is hydrolyzed at 170–230 °C, while cellulose undergoes only modest hydrolysis below 230 °C (Lü and Saka, 2010; Prado et al., 2016). Although higher temperatures release more C6 (glucose) sugars from the cellulose, they also promote a higher production of toxic compounds for the methanogenic microorganisms (Vallejos et al., 2012), for this reason we did not analyze temperatures higher than 200 °C. In addition, the hydrolysis at this temperature showed an excellent yield of sugars and low production of 5-HMFs and furfural inhibitors, where they could not be quantified, since they were detected below the minimum limit of the calibration curve.

Table 2
Products obtained from hydrolyzed with subcritical water at different temperatures.

Components (g.L ⁻¹)	Hydrolyzed	
	170 °C	200 °C
Galactose	0.1060 ± 0.0005	0.1304 ± 0.0039
Xylose	0.1059 ± 0.0034	0.2126 ± 0.0046
Mannose	nd	0.0812 ± 0.0011
Ethanol	nd	0.3020 ± 0.0051
Formic acid	nd	0.2799 ± 0.0014
Acetic acid	nd	0.2703 ± 0.0109
5-HMF ^a	nd	nd
Furfural ^a	nd	nd
pH	4.52	4.05
Residual Solid (%)	84.37 ± 1.35	45.72 ± 5.90

^a 5-HMF e furfural foram detectados abaixo do limite mínimo da curva de calibração (0.04 g.L⁻¹).

It was observed during the hydrolysis that the pH values of the two temperature conditions analyzed decreased due to the disruption of cell walls and the appearance of organic acids (Prado et al., 2016). The process temperatures had an important impact on the decay of the pH values, since in the experiment at 170 °C the pH of the first hydrolysis kinetic point was 5.01 and the last 4.29, resulting in the average pH of 4.52; while in the experiment at 200 °C, the pH went from 4.42 at the first collection point of the pretreatment to 3.95 in the last, and the average of the hydrolysate was 4.05. Therefore, with the temperature increase, a decrease occurs in the pH values due to the effect of higher production of organic acids. These organic acids react in the biomass structure, providing acidity to accelerate the hydrolysis reactions as an autocatalytic process (Cocero et al., 2018).

It was observed that percentage of solid residual remaining in the reactor after the subcritical hydrolysis of açai residue in the experiment at 170 °C was higher than the amount of solids in the experiment at 200 °C, indicating that in relation the initial amount of 100% açai residue, approximately 84% of this residue did not hydrolyze or partially hydrolyze. These results are also observed in the yields of reducing sugars, total reducing sugars and the monosaccharides, which were much lower when compared to the yields obtained in the hydrolysis in subcritical water at 200 °C. Thus higher hydrolysis and solubilization of hemicellulose is observed with increasing process temperature, as reported by Prado et al. (2014). However, the biomass remaining in the reactor was higher than those reported by Lachos-Perez et al. (2017); Torresmayanga et al. (2019), that is, even if the temperature condition at 200 °C was the best for açai hydrolysis, the percentage of residue remaining in the reactor (45.72%) was higher than that obtained in the subcritical water pretreatment of sugarcane bagasse and straw, probably the physical characteristics of the açai and the size of the particle used in this work.

Therefore, the subcritical water at 200 °C of temperature was selected as pretreatment condition for subsequent use in anaerobic digestion.

3.3. Anaerobic digestion

3.3.1. Solids content evolution

Fig. 3 shows the results obtained for solid (total, volatile, and fixed) during the anaerobic digestion process. Volatile solids represent organic matter and fixed solids to inorganic matter. Fig. 3 (A) shows the results of the integration reactor and Fig. 3 (B) shows the results of the control reactor.

The values of solids varied throughout the entire experiment for the integration reactor, obtaining 42.11 and 37.87% for total solids

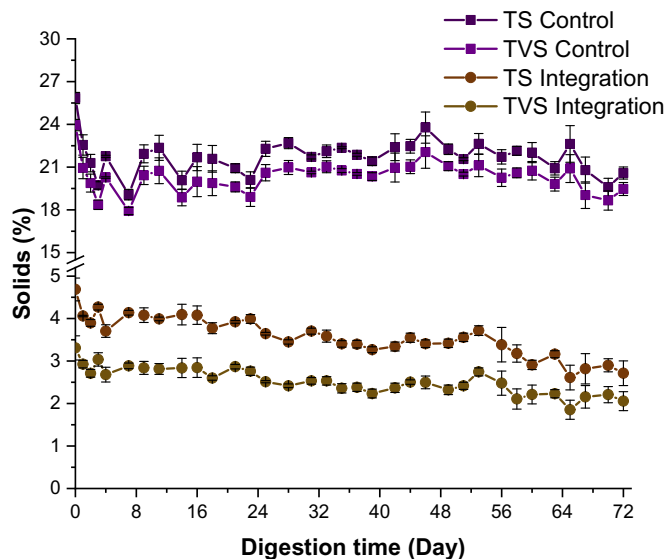


Fig. 3. Evolution of solids content during anaerobic digestion. (A) Integration reactor and (B) Control reactor.

and total volatile solids, respectively. These reductions occurred mainly due to the consumption of dissolved organic matter in the reactor hydrolyzed, due to the low solids content in this açai stone hydrolyzed. There was also a reduction in the values of total fixed solids. However, it is not related to the inorganic material consumption of the reactor, but to the increase in moisture that is caused by the ST reduction.

In the control reactor, the reduction of solids was accentuated during the initial 7 days, at the end of the first week of the experiment the decrease had been of 26.24% for TS and 25.18% for TVS. These results indicate a high microbial activity in the early stages of the anaerobic digestion process, where the bacteria hydrolyze more complex compounds, transforming them into simpler compounds, thus reducing the organic matter present in the system. After this initial week of the experiment the solids content showed small variations, the values remained relatively constant until the end of the digestion. In semi-continuous feeding processes involving microorganisms, this type of flotation occurs due to system limitations, feed overload, decreased hydrolytic activities of the microorganisms, with great experiments, and a series of factors such as pH, temperature and feed flow (Náthia-Neves et al., 2018).

3.3.2. Main operational parameters analysis

The evolution of the main operational parameters in the integration and control reactors along the anaerobic digestion are illustrated in Fig. 4. It can be observed in Fig. 4 (A) that during the first 10 days of the experiment both reactors presented pH with lower values than the remaining days. These results were observed because in this period there was no pH control with the intention of favoring the initial stages of anaerobic digestion, hydrolysis and acidogenesis phase (Kothari et al., 2014). The integration reactor reached 7.06 and the control reactor reached 5.14 in the first period. However, the decrease in pH values was more pronounced on the first day for the integration reactor, in the continuation, relatively constant pH values were observed. For the control reactor was reduction in the pH values until the end of the initial 10 days. After this first period, pH was maintained within the reactors between 7.0 and 8.0 to favor the subsequent steps of acetogenesis and methanogenesis (Kwietniewska and Tys, 2014). To this was added 6N sodium hydroxide (NaOH) to raise the pH to the desired range,

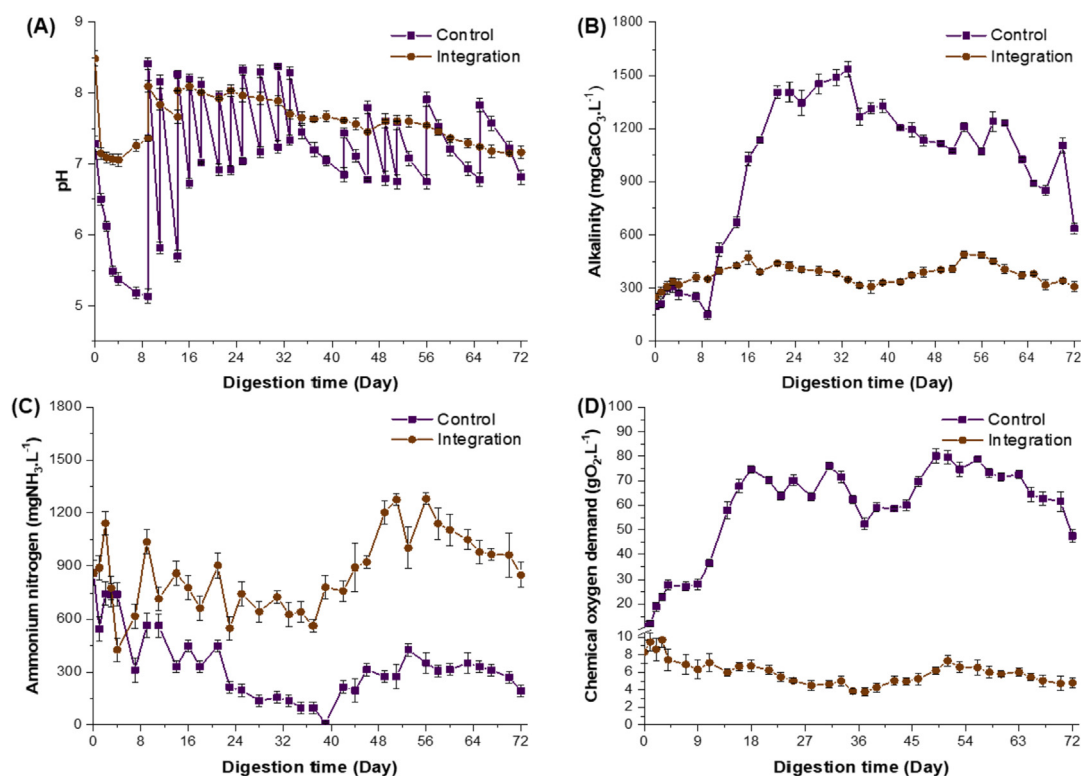


Fig. 4. Evolution of the main operating parameters along anaerobic digestion. (A) pH, (B) Chemical oxygen demand, (C) Alkalinity, and (D) Ammonium nitrogen.

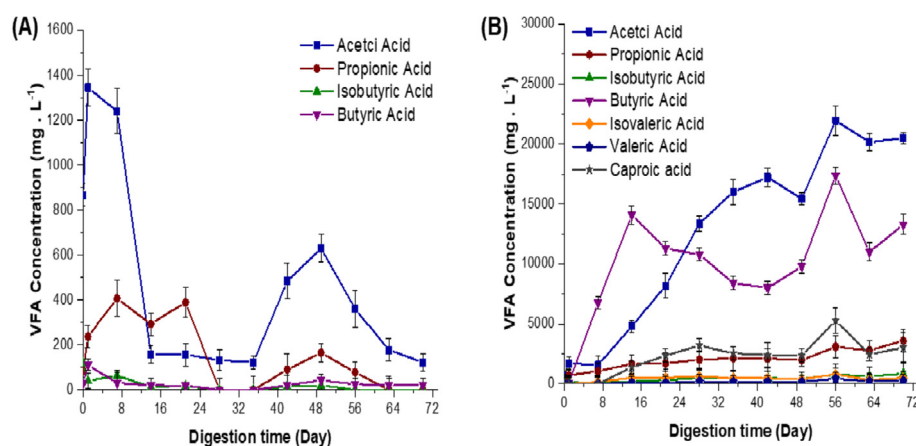


Fig. 5. Production of volatile fatty acids throughout the digestion time. (A) Integration reactor and (B) Control reactor.

and every time the pH fell back below that range. This control of pH was essential for the control reactor, due to the production of volatile fatty acids by the microorganisms from the lipids present in the açai biomass (Park et al., 2008).

The main reason for pH variations in this study was the difference in the composition of the substrates used in the different reactors. The reactor control contained a large amount of solid residues (acai seed). Therefore, the biodegradation was fast and, as consequence, the hydrolysis rate was higher and the pH decreased every day. Also, due to the significant concentration of lipids in the açai biomass, the control reactor produced a large amount of volatile fatty acids. The integration reactor produced less acids because the hydrolysate used as a substrate was free of lipids (see Fig. 5). On the contrary, the reactor with pre-treatment did not contain solid. Then, all the amount of food that was feed was biodegraded slower,

the formation of acids was not high, and the pH values were stable.

Alkalinity is a parameter related to the buffering capacity of anaerobic digestion systems, representing the ability of a system to neutralize weak acids. A low alkalinity solution will have its pH reduced by a weak acid, whereas a high alkalinity solution does not present a significant change in pH values since its free ions neutralize the weak acid (Palacios-Ruiz et al., 2008). Fig. 4 (B) illustrates the behavior of alkalinity during the anaerobic digestion process in the integration and control reactors. It is possible to observe that in the first days of the experiment, the alkalinity in the control reactor was relatively constant, but from the 10th (tenth) day it began to increase significantly, reaching a quintuple value. This increase in alkalinity values may have occurred due to the formation of carbonates and bicarbonates accelerated by the addition of sodium hydroxide used in pH control (Aceves-Lara et al.,

2012). On the other hand, the alkalinity in the reactor integration was practically constant throughout the experiment, since it was not necessary to do a lot of base addition for pH control in this reactor.

Referring to the results expressed in Fig. 4(A) and (B), we see a direct relationship between pH variations and alkalinity values in the control reactor. That is, with increasing alkalinity around the 20th (twentieth) day of digestion decreased pH variations caused by the production of volatile fatty acids. With this, we confirm the influence that alkalinity has on pH values in an anaerobic digestion process. For this and other reasons, alkalinity has been considered a better operational indicator than pH (Wang et al., 2018). A significant drop in pH value may occur when the anaerobic digestion process is severely unbalanced. However, alkalinity allows the detection of changes in the buffer capacity of the system. Thus, alkalinity monitoring could provide early warning of excessive accumulation of volatile fatty acids (Ward et al., 2011).

Ammonia is the end product of anaerobic digestion of proteins, urea and nucleic acids (González-Fernández and García-Encina, 2009). It is essential to be able to control the anaerobic process and to reduce the microbial activity (Zhang et al., 2014). Such an unstable situation can occur with ammonium nitrogen results at levels of 1,500–7,000 mgNH₃.L⁻¹, this wide range of inhibitory concentrations is due to differences in the nature of the substrates, inoculum, environmental conditions (temperature and pH) and acclimatization periods (Hejnfelt and Angelidaki, 2009). Fig. 4 (C) shows the evolution of the ammonium nitrogen content during the anaerobic digestion process of the integration and control reactors. Note that this parameter was profoundly affected by the temperature incident occurred on day 37 of the experiment, because in both reactors the ammonia concentration (NH₃) presented decreasing values. After the addition of mesophilic sludge to restore the rate of biogas production, there was an increase in the amount of ammonium nitrogen that returned to decay in the last days of digestion. These facts may have arisen because açai biomass contains little nitrogen and the sludge contains a higher amount (see Table 1).

In the integration reactor, the observed ammonium nitrogen content presented higher values than the control reactor, due to a probable hydrolysis/extraction of the proteins in the hydrothermal process of pre-treatment applied in the biomass of açai. In order to contribute to a better performance of this reactor, a reduction in the C/N ratio of the lignocellulosic feedstock of this work, according to Teixeira et al. (2005), was 48 C/N. Additionally, ammonia is related to alkalinity, as it increases the buffer capacity of the anaerobic digestion system, by neutralizing the excess volatile fatty acids formed and allowing a sufficient amount of these acids to produce biogas (Zhang et al., 2014). Thus also maintaining the pH of the reactor more stable integration, as observed in Fig. 4 (A).

The values of free ammonia are considered low for the control reactor because according to Rajagopal et al. (2013) a concentration below 500.0 mgNH₃.L⁻¹ can lead to biomass loss and reduction of biogas production yield due to lack of nitrogenous nutrients.

The chemical oxygen demand (COD) is an important parameter to verify the efficiency of the anaerobic digestion process concerning the biodegradability of organic matter (Naran et al., 2016). Fig. 4 (D) shows the evolution of soluble COD content during the anaerobic digestion process of the integration and control reactors.

We observed that for the integration reactor there was a 42% reduction in the COD value from the beginning to the end of the experiment. However, a variation occurred due to the addition of COD-rich mesophilic sludge (see Table 1) after day 37 superheating. The steadily declining value showed growth after this incident, rising from 3.77 gO₂.L⁻¹ on day 37 to 7.31 gO₂.L⁻¹ on day 51, but again dropping again from this day to the end of digestion.

In the first days of the anaerobic digestion of the control reactor, a significant increase in the COD value was observed, starting from 9.82 gO₂.L⁻¹ to 74.73 gO₂.L⁻¹ on day 18, the initial period in which conditions pH prioritized the phases of hydrolysis and acidogenesis. After the increase of COD probably caused hydrolysis and/or solubilization of the organic matter, we can observe a range with COD values relatively constant until the 31 day, from this day we notice a decay until day 37. Day in which the superheating occurred of the digestion system and there was an addition of inoculum that caused a new growth in the COD value, obtaining 81.12 gO₂.L⁻¹ at the 49th (forty-ninth) day, which soon afterward there was a decay up to 47.49 gO₂.L⁻¹ on the last day of the experiment. This represents a decrease of 41.44% in the COD value.

The açai seed is waste that has a large amount of lignocellulosic material in its composition. As a result of this complex material, with structure and hydrogen bonds that contribute to the low elimination of organic matter (Cho et al., 2011). The COD content at the end of the digestion was above the initial value, indicating the need for a more protracted process time to obtain a more significant removal of the organic material.

3.3.3. Production of volatile fatty acids (VFA)

The biochemical reactions that occur during anaerobic digestion promote the conversion of complex organic matter composed of carbohydrates, lipids and proteins into simpler substances such as monosaccharides, fatty acids and amino acids (Kothari et al., 2014). Fig. 5 shows the concentrations of volatile fatty acids produced during the anaerobic digestion assays: acetic (CH₃COOH), propionic (C₂H₅COOH), isobutyric (C₃H₇COOH), butyric (C₃H₇COOH), iso-valeric (C₄H₉COOH), valeric (C₄H₉COOH) and caproic C₅H₁₁COOH). Fig. 5 (A) illustrates the acids produced in the integration reactor and Fig. 5 (B) the acids produced in the control reactor.

The production of VFA through anaerobic digestion can be influenced by several factors, such as substrate nature, lipid concentration, hydraulic retention time (HRT), temperature or pH (Zhang et al., 2014). The amount of organic content to be hydrolyzed is the main factor responsible for the amount of VFA produced, together with the substrate composition (Khan et al., 2016). The temperature and pH have a direct influence on the solubility of the substrates, so the rate of hydrolysis and VFA production (Liu et al., 2012), can also control the type of VFA produced (Lee et al., 2014). TRH provides sufficient time for acidogenic bacteria to reduce residues in soluble derivatives and, consequently, favor VFA yield (Liu et al., 2018). Therefore, the nature of the substrate and the lipid concentration were the determining factors for the production of VFA in the reactor control having been so superior to the reactor integration.

Acetic acid was the main VFA produced in both reactors, also to the work of Bermúdez-penabad et al. (2017) who worked with tuna waste. The concentration of acetic acid in the integration reactor started high with 868.0 mg.L⁻¹, due to its high concentration in the açai hydrolyzed (see Table 2), increasing rapidly to 1345.0 mg.L⁻¹ remaining at this level for a week, shortly after declined dramatically to 158.0 mg.L⁻¹ remaining relatively constant until the thermal event of day 37, which promoted a new growth, reaching 629.0 mg.L⁻¹ on day 49 and ending with 122.0 mg.L⁻¹ at the end of digestion. The second most representative VFA was propionic acid, with a behavior similar to acetic acid. It had its highest production on the 7th (seventh) day of experiment with 406.0 mg.L⁻¹, keeping it moderately constant until day 21, then decaying and growing again after the addition of mesophilic sludge applied to recover the reactor. The other isobutyric and butyric acids were produced in low concentration, but the yield was partially constant throughout the experiment.

Although the acetic acid was the most concentrated VFA in the

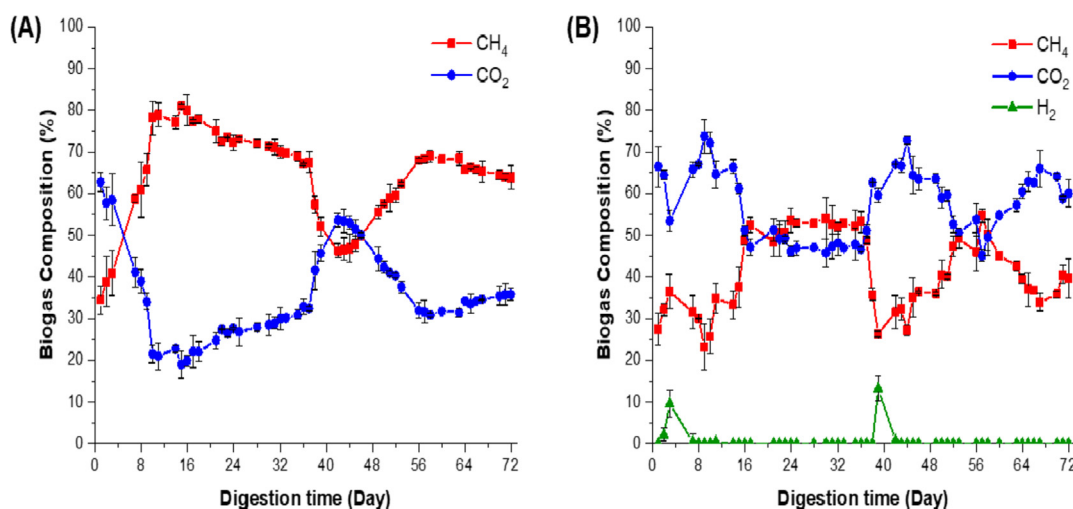


Fig. 6. Composition of the biogas produced. (A) Integration reactor and (B) Control reactor.

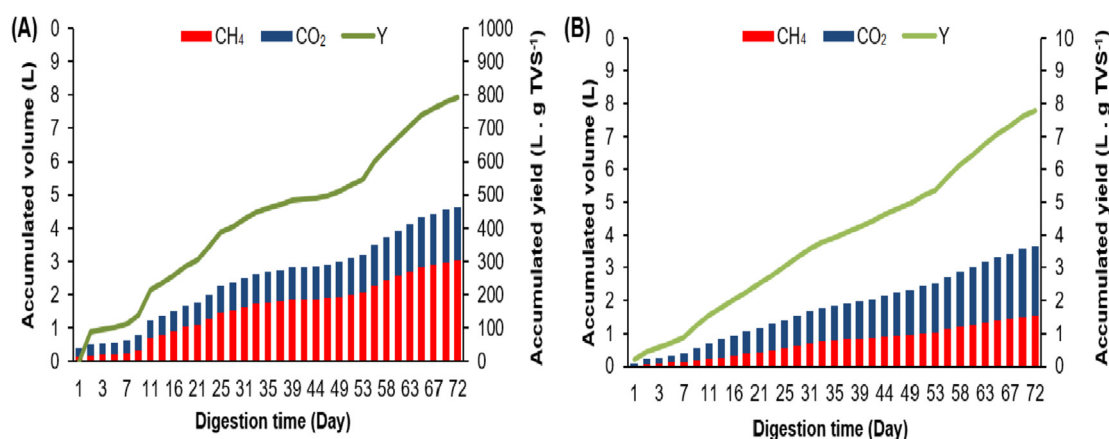


Fig. 7. Performance of accumulated volume and accumulated biogas yields along anaerobic digestion. (A) Integration reactor and (B) Control reactor.

control reactor, butyric acid had a high concentration, higher than acetic acid in some moments of digestion. Butyric acid was the most abundant VFA in the works of [Martinez-Jimenez et al. \(2017\)](#), [Náthia-Neves et al. \(2018\)](#), and [Pinto et al. \(2018\)](#). Due in no small amount of organic matter present in the control reactor, due to the biomass of açai, a significant change was not observed due to the addition of inoculum after the thermal incident of day 37. The higher production of the volatile fatty acids was obtained in 56th (Fifty-sixth) day of the experiment, with a concentration of the 4 most abundant acids, acetic acid, butyric acid, caproic acid and propionic acid equal to 22,961; 19,368; 5,289; and 3094.0 mg.L⁻¹, respectively. Isobutyric, isovaleric and valeric and butyric acids presented a small and constant production during the anaerobic digestion process. On the other hand, the higher concentration acids presented an increasing profile during the experiment time.

3.3.4. Biogas production

Fig. 6 shows the composition of the biogas produced in anaerobic digestion processes: methane (CH₄), hydrogen (H₂) and carbon dioxide (CO₂). The composition of the integration reactor is illustrated by Fig. 6 (A) and that of the control reactor by Fig. 6 (B).

Fig. 6 (A) shows the composition of the biogas produced by the integration reactor, which contained açai hydrolysate as a substrate, rich in reducing sugars fully accessible to the

microorganisms. Thus, the apex of methane production was rapidly reached, with only 10 days of the experiment, 78.23%. The maximum concentration of methane, 81.0% occurred on the 16th (sixteenth) day, after this day it is possible to observe a small linear decay in the methane concentration up to the 37th (thirty-seventh) day.

The decay in the concentration of methane for the reactor integration in the period from 16 to 37 and the day concentration was resumed until the end of the experiment at day 72, is related to a probable reduction in the activity of the methanogenic microorganisms involved at this stage of digestion, which tend to lose performance in their activities as digestion time passes, resulting in lower methane production. In addition, as mentioned previously, there was a reduction in the content of organic matter (substrate) throughout the digestion process, 37.87% in the total volatile solids content and 42% in the COD value.

It can be observed in Fig. 6 (B) that the methane concentration of the biogas produced in the control reactor was lower than that produced by the integration reactor since in this reactor the residual biomass of açai was used without any treatment with the nutrient source for the microorganisms.

There was a high production of hydrogen on the 3rd (third) day of digestion, reaching a concentration of 9.55%. Due to the high growth rates of acidogenic bacteria producing hydrogen ([Shen](#)

et al., 2016). The decrease in the composition of hydrogen in the biogas can be caused by the high hydrogen content, which makes the acidogenesis thermodynamically disadvantageous, which in turn favors the activity of the hydrogenotrophic bacteria that consume this hydrogen (Santos, 2013). After the hydrolysis and acidogenesis stages that were prioritized during the first 10 days of experiments, where the pH of the system was maintained between 5 and 6 (Fig. 4 (A)), methane production increased up to 16th (sixteenth) days reaching yield close to 50% in biogas. In this way, the methane apex was obtained 6 days after the reactor integration and with the difference of approximately 30% less of methane in the control reactor.

In this case, there was again a high production of hydrogen, 13.25% on day 39. In other words, we had a new start, promoted by the new microorganisms present in the process. The complete recovery of the methane composition occurred after 15 days of the thermal incident.

The volume and yield accumulated of biogas throughout the anaerobic digestion process are shown in Fig. 7. Fig. 7 (A) shows the results of the integration reactor and Fig. 7 (B) the results of the control reactor.

The accumulated volume of biogas was 3.71 and 4.64 L the biogas yield was 7.79 and 791.81 L.KgTVS⁻¹ for the control and integration reactor, respectively. The results show that pretreatment with subcritical hydrolysis of the açai residue was very efficient, increasing the biogas yield by more than 100 times. In addition to producing biogas richer in methane, as observed in Fig. 7.

Pinto et al. (2018) studied co-digestion of green coffee powder and vinasse, with 12% TS, produced 8.2 L during the 45 days of digestion and obtained a maximum accumulated yield of 18.0 L.KgTVS⁻¹. The substrate used by this work is difficult to digest also to the lignocellulosicaçai residue, however, it produced a higher volume of biogas and yield in relation to the volatile solids content added in the reactor, probably because it used a solids concentration well below 25% of TS used in the control reactor of this work. Besides, the results of the Triolo et al. (2012) indicate that substrates containing more than 100 g_{lignin}.kgTVS⁻¹ were directly related to methane production. For this work, 214 g_{lignin}.kgTVS⁻¹ was found in açai residue.

Náthia-Neves et al. (2018) worked with anaerobic digestion of food residue in methanogenic conditions, with 16% of initial TS for 77 days. They obtained 2.3 L of accumulated biogas volume and 311.41 L.KgTVS⁻¹ yields. A very high yield if we take into account the TS content, however, the digestibility of the food residue justifies this value. Which in turn is still well below the 791.81 L.KgTVS⁻¹ found in reactor integration, which utilized açai hydrolyzed.

4. Conclusions

The residual biomass of açai contains a total carbohydrate content close to 70%, which would justify the use of this residual biomass for the production of biomethane through anaerobic digestion technology. The subcritical water technology used as a pretreatment of the residual biomass of açai, proved to be efficient in the hydrolysis of sugars with low generation of inhibitory compounds. The experiments at temperature of 200 °C resulted in the best results, being this the condition selected for later application in anaerobic digestion.

Subsequently, the comparison of two anaerobic digestion reactors containing açai residues with pretreatment (integration) and control revealed that the control of pH in the range of methanogenesis was necessary in the control reactor due to the high production of volatile fatty acids (VFA), the ones with the highest

concentration being acetic and butyric acid. The reactor integration showed high buffering power, stable pH control and in the production of VFA stands out only acetic acid. Also, the integration reactor presented higher values of ammoniacal nitrogen than the control reactor, due to a probable hydrolysis/extraction of the proteins in the hydrothermal process of pretreatment applied in açai biomass.

The subcritical water pretreatment made the açai stone an excellent residual biomass for the production of biogas through anaerobic digestion, in view of the extremely high yield gain in the biogas production, 100 times greater when compared to the experiment without pretreatment, in addition to the high methane composition obtained, reaching 81% while reducing the organic load by approx. 45%.

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